Ravinder Singh

236-777-7674 | Gmail | LinkedIn | GitHub

Available: January 2027 – December 2027 (flexible to company requirements)

Education

Thompson Rivers University *Kamloops, BC*

Bachelor of Computing Science | *GPA: 3.64/4.33* | *Dean's List (3 Semesters)* Jan 2023 – Present

- Relevant Coursework: Computer Networks, Ethical Hacking, Operating Systems, Software Development, Software Engineering, Data Structures, Databases

Projects

HSE Compliance & Incident Tracking System | *React*, *Tailwind CSS*, *Node.js*, *Supabase* Sept 2026 – Present

- Leading a 4-member team to develop a Health, Safety and Environment (HSE) Compliance and Incident Tracking System for The Dawson Group, replacing fragmented Smartsheet workflows with a centralized web application.
- Developing responsive user interfaces and dashboard features using React and Tailwind CSS, including application navigation, shared layouts, and scorecard PDF export functionality.
- Coordinating weekly client meetings, sprint planning, GitHub project tracking, and project documentation while collaborating on system design and integration with Node.js and Supabase.

Real-Time Traffic Monitoring System | *React*, *TinyML*, *Firebase*, *Python* Sept 2025 – Nov 2025

- Developed a Raspberry Pi and camera-based system that tracks real-time passenger counts on buses and can be adapted to monitor traffic at any doorway.
- Why I built it: My Fall 2025 Internet of Things elective sparked my interest in using small, low-cost devices to solve everyday problems like tracking how full a bus is.
- What I learned: Linux command line, Raspberry Pi setup and OS installation, training TinyML models such as YOLO, and building a full-stack website with Firebase.
- Video Preview: https://www.youtube.com/watch?v=CXEYt82gjqI

Securing IoT Devices | *Network Security*, *TinyML*, *Firebase* Sept 2025 – Dec 2025

- Extending the Real-Time Traffic Monitoring System, developed a secure platform for data transfer between IoT devices and the cloud (Firebase) using rotating secure keys, rate limiting, and device blocking.
- Why I built it: While taking Network Security, I realized my traffic monitoring devices were easy targets, so I extended the project to protect them.
- What I learned: How to secure low-power IoT devices, often the most vulnerable entry points in a network, using rate limiting and rotating secure keys.
- Video Preview: https://www.youtube.com/watch?v=8OjqJszHSk4

Personal Portfolio Website | *React*, *React Router* May 2025 – June 2025

- Built a personal portfolio website showcasing my projects and skill set, with a place for visitors to view my work and connect with me.
- Why I built it: Inspired by other developers' portfolio sites, I used my summer break, while working 30 hours a week, to create my own.
- What I learned: Applied the React concepts I was studying, building a solid foundation in hooks, components, pages, and single-page routing with React Router.
- Direct Link: https://ravi200603.github.io/Portfolio/

Technical Skills

Languages & Frameworks: JavaScript, Python, Java, PHP, SQL, React, Tailwind CSS, Node.js
Development: REST API Integration (Front-End), Git & GitHub (Version Control), Technical Documentation
AI / ML: Machine Learning, YOLO Object Detection, TinyML
IoT & Cloud: IoT Device Integration, Cloud-IoT Monitoring, Firebase

Work Experience

Front Desk Agent Mar 2024 – Present

Howard Johnson *Kamloops, BC*

- Manage guest check-in and check-out processes while supporting daily front-desk operations.
- Perform end-of-day system audits using SynXis software to maintain accurate records.